8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

d) Flexible Manufacturing Systems (FMS)

Modular/cell production systems

Flexible Manufacturing Systems (FMS)

- ***A Flexible Manufacturing Systems (FMS)*** is the organisation of manufacturing to facilitate easily switching from one product to another in response to demand.

- It plays a very important role in facilitating Just in Time (JiT) production.

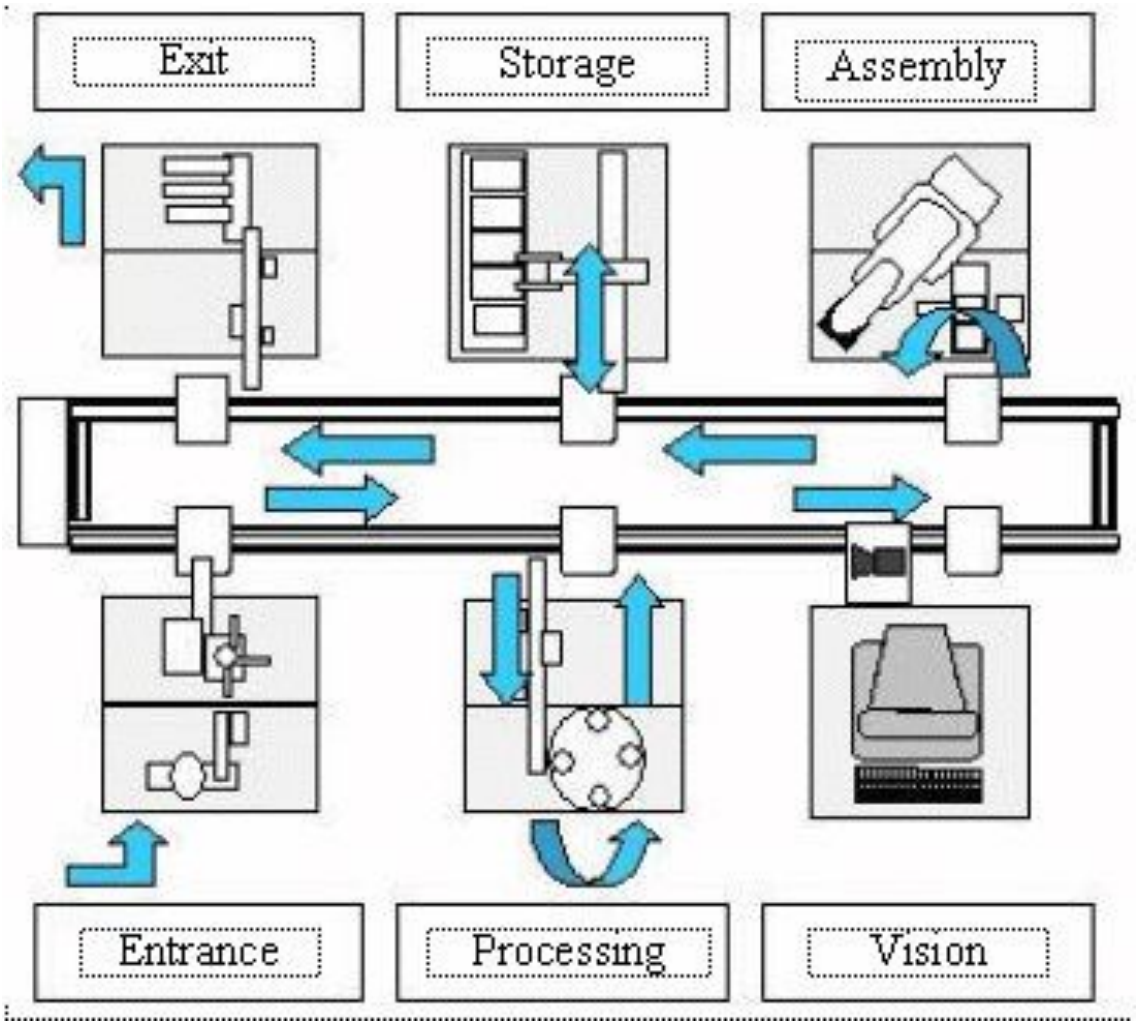
*There are **<u>two</u>** main features of flexible manufacturing systems….*

Flexible Manufacturing Systems (FMS)

1) Modular/Cell Production

- In **modular/cell production**, manufacturing cells combine CNC machines, such as lathes, milling machines, drills and grinders in a group – or 'cell'.
- The cell is programmed to carry out a sequence of operations to make parts such as car engine components.
- AGV's transport materials/parts to and from buffer zones.
- Robots with six axes (directions of movement) load and unload machines.
- Robots and AGV's are programmed along with the CNC machines they are servicing.

Diagram of a simple FMS Cell



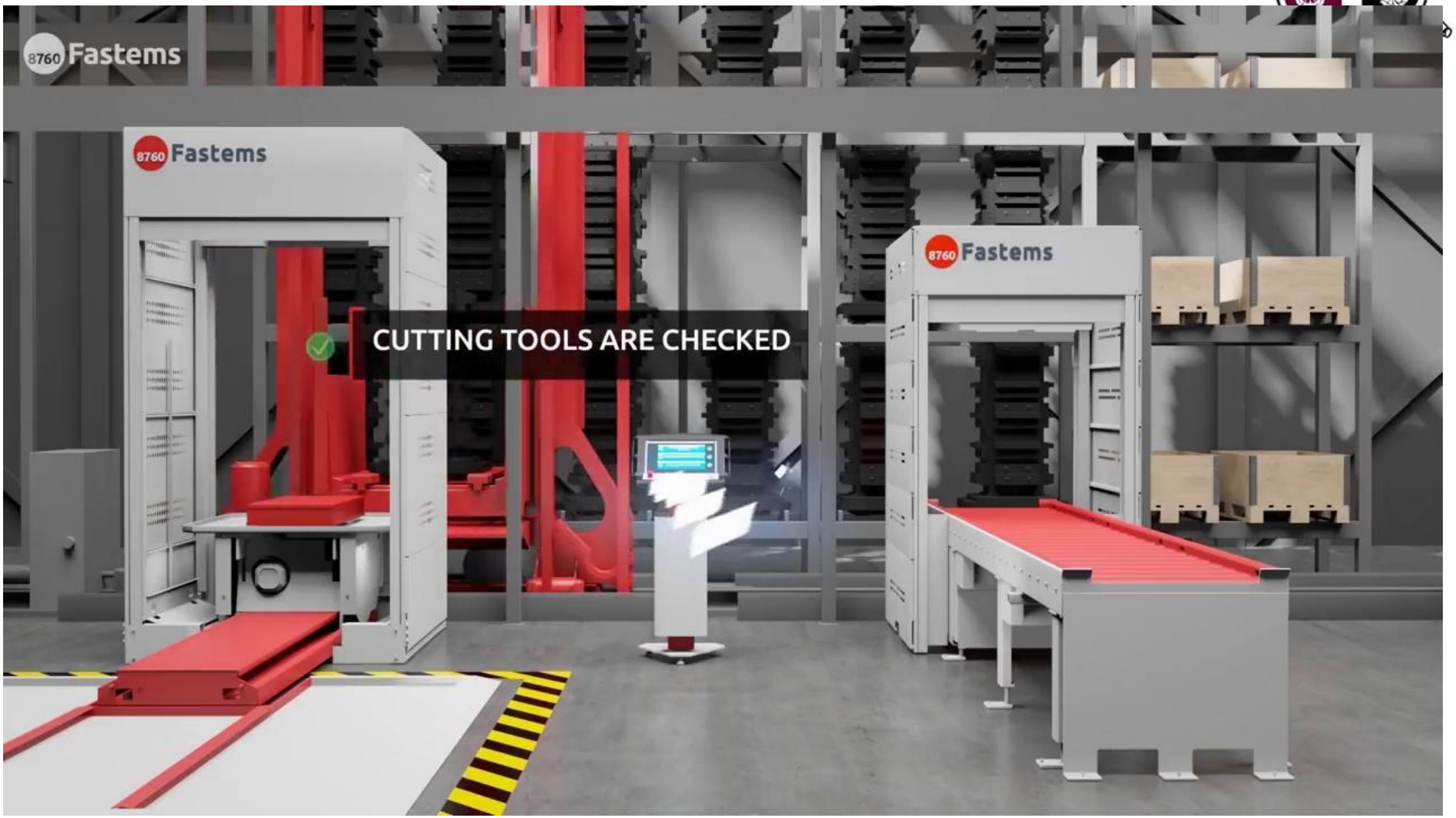
Flexible Manufacturing Systems (FMS)



2) Single Minute Exchange Dies (SMED)

- SMED means changing dies/moulds in machines in a target time of under ten minutes.
- Involves quick-release locking devices and multi-purpose, interchangeable elements.
- Examples include changing dies for pressed sheet metal car radiators and various injection-moulded parts.
- Before SMED, machines were idle for many hours during mould/die changes (downtime).
- Minimum downtime is required to obtain the maximum possible revenue from investment in machines.

Video: Flexible manufacturing systems (FMS)



Flexible manufacturing systems (FMS)

Link to alternative video:

https://www.youtube.com/embed/YGtg4OPSFhc

Flexible Manufacturing Systems (FMS)

Additional information….

Flexible manufacturing systems (FMS)

- FMS is a form of flexible automation, several machines are linked together.
- All aspects of FMS are controlled by a central computer.
- FMS brings together new manufacturing technologies eg. CNC machines, robotics, automated material handling, etc. to form an integrated system.
- It is different from an automated production line due to its ability to process more than one product style simultaneously.
- FMS systems have powerful computing capacities which allow it not to only control machinery but also to perform production planning and routing of materials through the system.

Advantages
▪ Increased productivity due to automation. ▪ Shorter lead times for new products due to flexibility. ▪ Lower labour costs due to automation. ▪ Improved production quality due to automation.

Disadvantage
▪ Setting up the system requires a great deal of pre-planning and very high set up costs.

Flexible manufacturing systems (FMS)

- FMS vary in their size and complexity.
- Some are designed to be very flexible and to produce a wide variety of parts in very small batches.
- Others have the ability to produce a single complete product in large batches from a sequence of many individual operations known as a flexible TRANSFER LINE

FMS Systems:

- Can produce a variety of the part
- Can adapt to a change in schedule and volume.
- Breakdown of part of the system does not stop all manufacture.
- A new part design can be introduced relatively easily.

Advantages
▪ Increased productivity due to automation. ▪ Shorter lead times for new products due to flexibility. ▪ Lower labour costs due to automation. ▪ Improved production quality due to automation.

Disadvantage
▪ Setting up the system requires a great deal of pre-planning and very high set up costs.

MACHINE FLEXIBILITY:

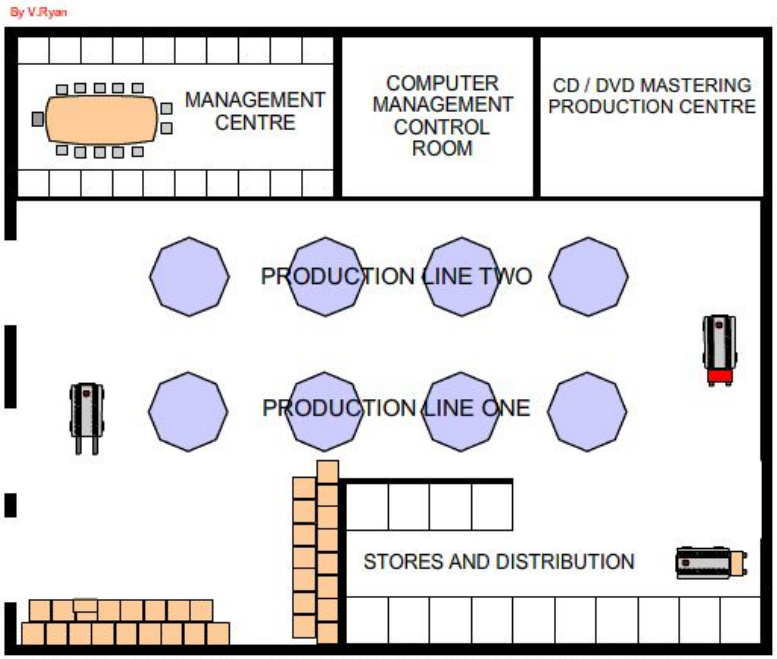
A Flexible Manufacturing System is one that can be changed or adapted rapidly to manufacture different products or components at different volumes of production. Flexible manufacturing systems are usually seen at their most efficient when manufacturing components rather than finished products.

When machines are used to manufacture a variety of products/components this is called ‘machine flexibility’. A good example is a company that specialises in manufacturing injection moulded components. The component design may change very quickly but as long as the mould is updated / modified the injection moulding machine can still operate. Computers monitor the supply of materials, monitor the operating temperatures and quality control. Machines of this nature are adaptable and are ideal for FMS.

1) Modular/Cell Production - ROUTING FLEXIBILITY:

When the manufacturing system is flexible enough to absorb changes in manufacturing capacity such as an increase in the number of units being made this is called 'routing flexibility'. This may be achieved by selecting the right path through which the product or component flows through the production line. A production line that has enough manufacturing capacity that can be utilised quickly is one that has routing flexibility. For example a CD/DVD manufacturing facility may have a number of manufacturing units in standby mode until their capacity is required. When not required units can be shut down rapidly. This makes a manufacturing plant more efficient.

The example below shows a CD/DVD manufacturing plant. It has two production lines. Each production line is automated and can be activated very quickly. For example, if there is a need to increase the manufacture of a new music CD, both production lines can be used. Also, if there is a need to speed up any stage of manufacture, for instance the screen printing of each completed CD - then the screen printing bay of each production line can be used. This system has built in flexibility.



AN IMPORTANT ASPECT OF FLEXIBLE MANUFACTURING SYSTEMS IS COMPUTER CONTROL

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2) SMED - MACHINE FLEXIBILITY:

The products below are CD /DVD storage units. The injection moulding machine is an example of machine flexibility because the mould can be changed quickly and different designs can be manufactured.

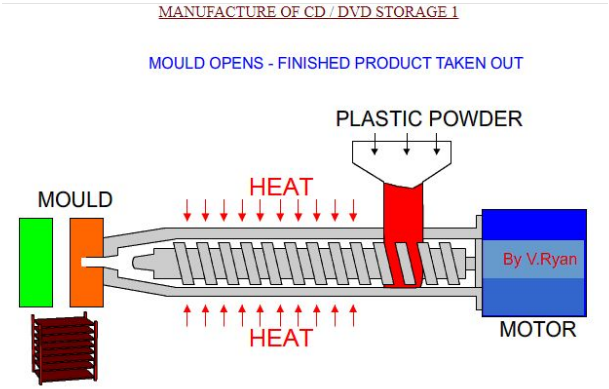
CD / DVD STORAGE 1



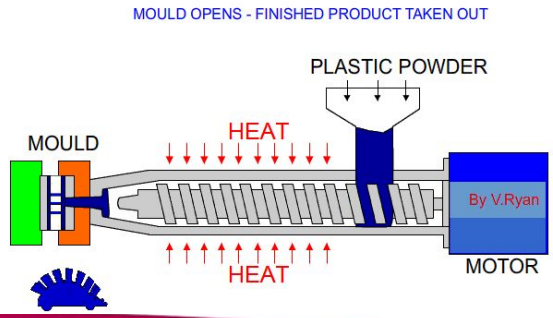
CD / DVD STORAGE 2



MANUFACTURE OF CD / DVD STORAGE 1



MANUFACTURE OF CD / DVD STORAGE 2



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Now Test Yourself:

1. What do the following terms mean and why are they both important in servicing ***flexible manufacturing cells (FMC's)***?
   - ***Automated Storage and Retrieval Systems (ASRS)***
   - ***Automated Guided Vehicles (AGV)***

*** Look at and refer to sections *c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* and *d) flexible manufacturing systems (FMS), modular/cell production systems****